

Retrieval-Augmented Generation Architecture Overview

Introduction to RAG

Retrieval-Augmented Generation (RAG) merges parametric knowledge from large language models with non-parametric knowledge from external document stores. By querying external indexes, RAG mitigates hallucinations and enables real-time domain adaptation without expensive fine-tuning.

Vector Indexing & Embeddings

Dense vector indexing relies on embedding models such as BERT, RoBERTa, or OpenAI text-embedding-3 to project documents into low-dimensional vector spaces. Nearby vectors correspond to semantically similar texts, enabling nearest-neighbor retrieval via HNSW or IVF indexes.

Context Assembly & Grounding

Retrieved document chunks are injected into the LLM prompt context window. Grounding constraints force the model to cite specific sources and restrict response generation strictly to facts present in the retrieved context.